Additional Composition Pair: Instinct/Reasoning Separation and FAI Exchange Bounding

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Attribution

> This work derives from and formalizes the inter-Self coordination architecture introduced in "Inter-Self Coordination via Shared Substrate / Full Aspect Integration" (Li, April 2026), the third paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026) and "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026).

Abstract

> Two architectural commitments govern FAI exchange content: exchange bounding (D1.10), which restricts FAI exchange to DNA-layer and action-layer substrate content, and instinct/reasoning separation extended to inter-Self scope (D2.40), which specifies why the instinct layer — including the substrate harness as well as LLM weights — must remain within the home perimeter during FAI events. Considered separately, each commitment performs limited governance work. Considered together, they produce three non-obvious governance requirements that neither commitment generates alone: a theoretically grounded bounding test that closes the gap between a list of prohibited content types and a principled violation criterion; a specific detection obligation for harness substrate content as a distinct and often-overlooked violation category; and an explicit articulation of bounding as governance sovereignty protection over each Self's organizational character. This note formalizes the composition and derives its governance requirements.

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1. Pair Identification

Commitment A (D2.40) — Instinct/reasoning separation at inter-Self scope. At the intra-Self scope established by Paper 2, the instinct layer (the substrate harness and associated LLM) is structurally separated from the reasoning layer (DNA-layer and action-layer substrate content). Paper 3 extends this separation to inter-Self scope: during a FAI event, the instinct layer of each participating Self remains within that Self's home perimeter. Only reasoning-layer content — DNA-layer governance specifications and action-layer operational records — participates in the exchange through the shared substrate. This extension is not incidental; it carries the instinct/reasoning separation as a foundational trilogy commitment into the inter-Self coordination architecture without modification.

Commitment B (D1.10) — FAI exchange bounding. The FAI exchange is bounded to substrate content: DNA-layer content (orchestration substrates, behavior specifications, schemas, governance rules) and action-layer content (recorded task instances, operational outputs, lived experience) may exchange across the inter-Self perimeter through the shared substrate. LLM weights and instinct-layer content must not cross the perimeter. This commitment specifies the boundaries of the exchange operation as an architectural constraint on FAI as a mechanism.

The composition scenario is an active FAI event in which both commitments apply simultaneously. Governance must verify that the exchange bounding (Commitment B) is satisfied — that the exchange contains only permitted content types — while also accounting for the instinct/reasoning separation (Commitment A) as the theoretical basis for what makes a content type permitted or prohibited. Both commitments are active, and neither produces the full governance requirement without the other.

2. Governance Scenario

A FAI event is operating between two Selves. The shared substrate is accumulating aspects from both participating Selves. Exchange bounding verification is required: a governance observer must confirm that the content participating in the exchange is bounded to DNA-layer and action-layer substrate content, and that no prohibited content has crossed either perimeter.

The governance task appears straightforward under Commitment B alone: check that no LLM weights are present in the shared substrate. But this reading is incomplete. The instinct layer is not limited to LLM weights. It also includes harness substrate content — the governance specifications and infrastructure configuration that govern how each Self's LLM connects to and operates within the substrate. Harness substrate content is not LLM parameter data, and it is governance documentation, but it governs instinct-layer infrastructure rather than reasoning-layer direction. A bounding check that stops at LLM weights fails to detect harness substrate migration.

Without Commitment A, there is no architectural basis for classifying harness substrate content as a violation category distinct from LLM weights, and no theoretical grounding for why the violation boundary falls where it does. Commitment A supplies both. The composition produces a verification task that is more specific, and more theoretically defensible, than either commitment generates independently.

3. Non-Obvious Governance Requirements

Requirement 1 — Theoretical grounding makes the bounding test precise. D1.10 specifies what is bounded: exchange is restricted to DNA-layer and action-layer content. D2.40 explains why: the instinct layer is where each Self's organizational character resides; allowing instinct-layer content to cross perimeters would allow one Self's fundamental operational dispositions to influence another's. The composition transforms the bounding check from a content-type list into a principled violation criterion: content that crosses the inter-Self perimeter is bounded-compliant if and only if it is (a) DNA-layer governance specifications or (b) action-layer operational records. It is a bounding violation if it is (c) harness substrate content governing instinct-layer infrastructure, (d) LLM weight representations, or (e) any content that would influence how the receiving Self's LLM operates at the infrastructure level — not merely what it is directed to do by governance.

The gap this closes is important. Without theoretical grounding, the bounding rule "no LLM weights" is a specific prohibition that leaves open whether other instinct-layer content is similarly prohibited. With D2.40's theoretical grounding, the rule generalizes: the prohibition follows from the principle that instinct-layer influence must not cross perimeters, and any content that would carry such influence is prohibited regardless of its specific form. Governance practitioners applying the theoretically grounded test can classify novel content types by asking whether they carry instinct-layer influence rather than by maintaining an enumerated exclusion list.

Requirement 2 — Harness substrate content requires specific detection in bounding verification. Exchange bounding verification (as formalized at D2.16) must distinguish DNA-layer orchestration rules — which are permitted to exchange — from harness substrate content — which is not. The distinction is subtle: both are governance documents about LLM behavior, and both may appear in substrate form. The difference is functional. DNA-layer orchestration rules specify what the LLM is directed to do at the reasoning layer: its tasks, its coordination responsibilities, its decision authorities. Harness substrate content specifies how the LLM connects to and operates within the substrate infrastructure at the instinct layer: its connectivity, its execution environment, its infrastructure-level configuration.

The composition requires that D2.16 verification be specific about harness substrate content as a distinct violation category, not merely as an example of the general LLM-weights prohibition. A governance check that examines shared substrate content for LLM parameter data, confirms it is absent, and proceeds to approval is incomplete. Harness substrate content may be present in non-parametric form — as configuration files, infrastructure specifications, or execution environment documents — and still constitute a bounding violation. Governance practitioners must be trained on the harness-vs-DNA distinction specifically for the bounding verification check to be meaningful.

Requirement 3 — Exchange bounding protects governance sovereignty over organizational character. The composition reveals what is at stake in the bounding commitment. The instinct layer is not simply a technical implementation layer that happens to be excluded from exchange for engineering convenience. It is the layer where each Self's fundamental operational character is expressed: the dispositions that govern how the Self operates at the substrate infrastructure level, how its LLM connects to its coordination environment, and what its baseline operational profile looks like before any reasoning-layer direction is applied.

If instinct-layer content crosses perimeters through the shared substrate, one Self's fundamental character could influence another's through the exchange event. This is not a data-contamination risk; it is a governance sovereignty risk. Each Self's governance authority over its own operational dispositions — its ability to determine what kind of entity it is at the infrastructure level — would be compromised by unauthorized instinct-layer influence from a coordination partner. Exchange bounding is not a technical data filter; it is the architectural mechanism that protects each Self's governance sovereignty over its own character. This sovereignty stake is why the commitment is architecturally non-negotiable rather than merely recommended.

4. Prior-Art Significance

The three properties identified above — theoretically grounded bounding test, harness-vs-DNA distinction for verification, and bounding as governance sovereignty protection — are specific to this composition pair. They do not appear in either commitment independently, and they are not derivable from either commitment without the other.

Any architecture that implements governed inter-AI coordination with boundary enforcement at the inter-Self perimeter must address all three. An architecture that enforces exchange bounding without theoretical grounding produces a bounding test that is a content-type list rather than a principled criterion; it cannot systematically classify novel content types. An architecture that checks for LLM weight transfer but does not distinguish harness substrate content from DNA-layer content misses the subtler violation category. An architecture that treats bounding as a technical data filter rather than as a sovereignty protection may be configured to relax the commitment where governance considers it convenient, undermining the non-negotiable character of the architectural constraint.

Prior art that fails to address any of these three properties is not equivalent to an implementation that addresses all three, even if it satisfies the surface-level requirement that instinct-layer content not cross perimeters.

5. Operational Test

For exchange bounding verification, an observer can confirm the composition is operationally instantiated by verifying the following:

(a) Harness substrate content is named as a specific violation category. The verification protocol explicitly identifies harness substrate content — specifications governing how each

Self's LLM connects to and operates within the substrate infrastructure — as a prohibited content type distinct from LLM weights. The check is not satisfied by confirming LLM weights are absent; it requires a separate classification step for harness substrate content.

(b) DNA-layer orchestration rules and harness substrate content are classified separately. The content-type review in bounding verification treats DNA-layer orchestration rules (reasoning-layer direction; permitted to exchange) and harness substrate content (instinct-layer infrastructure; not permitted to exchange) as separate categories. Documentation that is governance-relevant and concerns LLM behavior is not automatically classified as DNA-layer content without this distinction being applied.

(c) The governance record explicitly references instinct/reasoning separation as the theoretical basis for violation classification. The verification record for exchange bounding does not merely state that prohibited content types are absent. It records that the classification of prohibited content types follows from the instinct/reasoning separation commitment: content is prohibited if it would give one Self influence over another's instinct-layer operations, regardless of the specific form in which that influence would be exercised.

A bounding verification protocol that satisfies all three criteria instantiates the composition. A protocol that satisfies D1.10 alone — confirming that exchange is restricted to substrate content without applying the theoretically grounded violation criterion, without distinguishing harness from DNA content, and without grounding the classification in the sovereignty rationale — satisfies the sub-commitment but not the composition.

Source Papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems.* April 2026. ORCID: 0009-0004-8065-3235.

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